Role-Based Working & Complete Operational Responsibilities Guide

The Authoritative Operational Standard for B2B FMCG Distribution — From Core Mandates to Micro Possibilities

Target Platform: SupplySetu AI-Powered B2B Distribution SaaS	**Client Operational Cluster:** Jabalpur FMCG Hub (Madhya Pradesh, India)
Network Scope: 1,800+ Kirana Retailers & 18 Dedicated Sales Beats	**Daily Order Throughput:** 600+ B2B FMCG Invoices / Rs 85-100 Cr GMV
Integrated Tech: React + Node/TS + Supabase + Dexie.js + Gemini AI	**Document Classification:** Production Role Architecture Standard

1. Executive Summary & Operational Mandate

SupplySetu is an enterprise-grade B2B distribution management and intelligent order orchestration platform specifically engineered for Indian FMCG wholesale distribution ecosystems. In traditional trade networks such as Jabalpur, distributors manage 1,800+ independent kirana shops using heavily fragmented, manual communication channels—unstructured WhatsApp messages written in Hindi and Hinglish, photographed notebook pages, and verbal phone orders. This conventional mode causes chronic operational friction: frequent SKU misidentification, mismatched packaging units (cases vs. individual packs), out-of-stock ordering, unapplied trade schemes, and days of manual data entry into accounting systems like Tally ERP.

This official document establishes the **definitive role-based operating model** of SupplySetu. It specifies the boundaries, primary workflows, secondary tasks, and micro-level edge-case responsibilities for each human and automated role: **Kirana Retailer**, **Beat Salesman**, **Distributor Order Desk Admin**, and the **Automated System Engine**. Every possible scenario—from low-connectivity beat visits and offline idempotency to multi-tier dealer pricing and Tally RFC-4180 batch reconciliation—is systematically defined to ensure complete organizational clarity and zero operational ambiguity.

1.1 Key Business Transformation Objectives

• **Digital Capture Rate:** Transition >50% of monthly ordering volume to digital and AI-assisted channels within 120 days of deployment.
• **Error Elimination:** Slash order mismatch and return rates from ~5% down to under 0.8% through real-time stock validation and AI confirmation screens.
• **Processing Velocity:** Compress average order turnaround time from 18 hours (next-morning Tally entry) to under 30 minutes for picking and dispatch.
• **Financial Discipline:** Enforce real-time credit checks, eliminating unauthorized over-limit supply and automating khata collection follow-ups.

CORE ARCHITECTURAL GOVERNANCE PRINCIPLE: SupplySetu enforces strict multi-tenant role isolation across the entire stack. Under no circumstance may a Retailer view competitor prices, credit profiles, or distributor queues. Beat Salesmen operate with retailer-delegated booking scopes strictly bounded by their daily beat itinerary. The Distributor Admin holds full fiscal, dispatch, and master catalogue sovereignty.

SECTION 2 | MASTER ROLE-BASED ACCESS CONTROL (RBAC) & SCOPE MATRIX

Comparative Analysis of Capabilities, Interfaces, Data Visibility, and System Authority

The matrix below contrasts the operational boundaries, data mutability, interface responsibilities, and system authority across all three primary human actors and the underlying automated system engine:

Dimension / Capability	Kirana Retailer	Beat Salesman	Distributor Admin	System / AI Engine
Core Operational Mission	Self-service order booking, price discovery & khata balance tracking.	Route field coverage, in-person booking & zero-network offline sync.	Centralized order triage, inventory control, credit approval & ERP export.	Autonomous NLP parsing, rate card calculation & data synchronization.
Primary UI Channel	Retailer Mobile/Desktop Portal + AI Order Pad Modal.	Mobile Beat Pad + Dexie Offline Order Pad + Queue Modal.	12-Column Live Order Desk + Financials + CRM + Tally Portal.	Headless REST API + WebSocket Hub + Background Workers.
Pricing & Schemes Visibility	Assigned Tier Dealer Rate (Standard, Tier 1, 2, Wholesale) & active deals.	Retailer-specific rates for assigned beat stores; deals catalogue.	Full master rate matrix (Base, MRP, Tier discounts, Scheme rules).	Programmatic access to rate cards, schemes, and tax slabs.
Order Placement Scope	Self-orders only. Restricted to personal merchant account.	On-behalf orders for assigned beat merchants only.	Manual order creation, modification, approval, or rejection.	Validates payloads; creates immutable order records in database.
Offline Operation Mode	Read-only browsing cached; requires connection to submit.	**100% Full Offline Punch** via IndexedDB; background auto-sync.	Online enterprise portal connected to live PostgreSQL.	IndexedDB storage layer on client; atomic sync on server.
Credit & Khata Authority	View own limit, balance, utilization; simulate UPI payments.	View assigned store balance before visiting; soft collection reminders.	Adjust credit limits, review holds, override hard stops, audit ledgers.	Enforces credit checks; flags CREDIT_HOLD when order > limit.
Inventory / SKU Mutability	Strictly read-only; sees stock badges (In Stock / Out of Stock).	Read-only cached catalogue; real-time stock alert on sync.	Full CRUD: Add SKUs, adjust physical stock, update MRP/HSN/GST.	Decrements available stock upon order dispatch transition.
ERP / Accounting Link	Download/Print GST Tax Invoices for internal accounting.	None (focus on field throughput and route execution).	1-Click Streaming RFC-4180 CSV export formatted for Tally ERP.	Sanitizes and serializes approved orders into RFC-4180 CSV stream.
Dispatch & Logistics	View real-time delivery status (Pending, Dispatched, Delivered).	View delivery commitments and route dispatch schedules.	Print picking sheets, assign delivery tempo, mark delivered.	Dispatches WebSocket/SSE delivery alerts to mobile clients.
AI Assistance Scope	Paste raw Hinglish notes; verify parsed cart items.	Quick search and voice-to-text cart population.	Review AI confidence scores and manual mapping overrides.	Executes Gemini API extraction with fuzzy phonetic fallbacks.

SECURITY & ANTI-TAMPERING ENFORCEMENT PROTOCOL: Role boundaries are hardcoded in server-side Express middleware (middleware/auth.ts) using JWT role claims. Client-side route protections exist solely for user experience. Any attempt by a client to submit unearned dealer prices, override credit holds, or access unassigned retailer records is trapped and rejected by server-side Prisma ORM queries.

SECTION 3 | ROLE SPECIFICATION: KIRANA RETAILER (MERCHANT PARTNER)

Self-Service FMCG Procurement, Multilingual AI Ordering & Credit Management

3.1 Role Overview & Operational Mandate

The **Kirana Retailer** represents the vital customer endpoint of the SupplySetu network (e.g. Anil Patel of Shri Ganesh Kirana Store, Lordganj). Retailers are independent proprietary grocery owners seeking rapid inventory restocking, transparent tier rates, automated trade promotions, and reliable credit tracking without waiting for physical salesman visits.

3.2 Primary (Macro) Responsibilities

• **Catalogue Discovery & Live Stock Auditing:** Browse authorized FMCG inventory across packaged foods and home care. Check live stock indicators before ordering to eliminate customer-facing stockouts.

• **Multi-Modal Digital Order Placement:** Construct purchase orders via the visual digital catalogue cart or by pasting unstructured WhatsApp messages into the AI Order Pad.

• **AI Order Verification & Cart Confirmation:** Act as Human-in-the-Loop reviewer: inspecting AI-matched SKUs, adjusting pack quantities, resolving brand ambiguities, and accepting stock alerts.

• **Automated Trade Scheme Maximization:** Leverage dynamic schemes ('Buy 10 Cases Get 1 Free', volume percentage discounts) applied in the cart to maximize retail resale margins.

• **Khata & Credit Limit Governance:** Monitor outstanding trade credit against approved distributor limits (e.g. Rs 75,000 ceiling), track available headroom, and resolve credit hold situations.

• **Order Tracking & Tax Invoice Reconciliation:** Track order status progression (Pending -> Approved -> Dispatched -> Delivered). Download official B2B Tax Invoices with GST breakdowns for Input Tax Credit (ITC).

3.3 Granular Micro-Responsibilities & Subtle Operational Possibilities (Small-to-Small Details)

Operational Event / Feature	Granular Responsibility & System Behavior	Edge Case / Failure Prevention
Hinglish & Dialect AI Input	Retailer pastes informal text (e.g., *'bhaiya 5 peti biscuit chhota, 2 bori detergent 1kg'*). Retailer must review parsed quantities and units before confirmation.	If Gemini AI hallucinates or maps an incorrect pack size (80g vs 120g), retailer adjusts via dropdown before adding to cart.
Tiered Pricing Margin Awareness	Retailers are assigned to pricing tiers (STANDARD 0%, TIER_1 2%, TIER_2 4%, WHOLESALE 6.5%). Retailer must check dealer rate vs MRP to maintain healthy margins.	Rates are filtered server-side; retailers cannot inspect competitor pricing tiers or negotiate unearned rates.
Free Product Scheme Multipliers	When ordering qualifying quantities (e.g. 25 cases on 'Buy 10 Get 1 Free'), retailer must recognize schemes use integer floor division (25 yields 2 free cases, with 5 remainder).	Cart summary explicitly displays free units as separate Rs 0.00 line items with tax-exempt treatment.
Credit Hold Exceedance Protocol	If order total pushes balance beyond credit limit (e.g. Rs 22,000 order with Rs 15,000 available headroom), order enters **CREDIT_HOLD** status upon placement.	Retailer can reduce order size or initiate an instant simulated UPI payment to immediately unlock credit headroom.
UPI QR Payment Simulation	Retailer enters payment amount on Khata screen, scans simulated dynamic QR, and submits reference. Instant credit reflection unlocks pending orders.	Submitting a payment amount greater than outstanding debt is rejected with a validation warning.
1-Click Repeat Order Mechanics	Retailer can duplicate previous orders with a single tap. System checks current warehouse stock for each item before populating the cart.	Any discontinued or out-of-stock SKUs are automatically pruned and flagged in a prominent warning toast.
Wholesale Unit & MOQ Multiples	Each SKU enforces wholesale box multiples (e.g. 1 BOX = 72 packets). Retailers cannot place orders for fractional cases or loose packets.	Quantity spinners enforce MOQ steps automatically; typing invalid numbers snaps to the nearest valid case multiple.

SECTION 4 | ROLE SPECIFICATION: BEAT SALESMAN (FIELD SALES EXECUTIVE)

Daily Route Execution, Zero-Network Offline Order Punching & Khata Collections

4.1 Role Overview & Operational Mandate

The **Beat Salesman** (e.g. Vikram Sharma, Beat #4 Lordganj & Sarafa) is the distributor's mobile field force. Salesmen visit 30-40 retail stores daily, punch on-behalf orders for non-tech-savvy merchants, conduct shelf audits, and monitor payment discipline. In dense bazars with severe network dead-zones, salesmen rely on a **100% offline-first Dexie.js IndexedDB order pad**.

4.2 Primary (Macro) Responsibilities

• **Beat Route Sequence Execution:** Follow daily assigned beat sequence. Visit mapped stores, log visit statuses ('Visited' vs 'Pending'), and maintain disciplined route pacing.

• **On-Behalf Order Punching for Kiranas:** Book commercial FMCG orders on behalf of kirana proprietors directly onto the mobile pad, selecting case quantities, confirming schemes, and noting merchant requests.

• **Pre-Order Khata & Overdue Auditing:** Inspect retailer debt and available credit headroom before entering the store. Flag overdue balances and negotiate payment promises before booking new stock.

• **Zero-Network Offline Order Booking:** Operate smoothly in complete network absence using local Dexie.js IndexedDB storage without UI lag, spinner timeouts, or data corruption.

• **Idempotent Synchronization Management:** Ensure all queued offline orders are successfully transmitted to the Jabalpur distribution cloud upon network restoration, verifying zero duplicate bookings.

• **Sales Targets & Performance Tracking:** Track daily Gross Merchandise Value (GMV), productive call counts (orders booked vs total visits), and monitor monthly incentive progress.

4.3 Granular Micro-Responsibilities & Subtle Operational Possibilities (Small-to-Small Details)

Operational Event / Feature	Granular Responsibility & System Behavior	Edge Case / Failure Prevention
Morning Catalogue Cache Priming	Before leaving the central warehouse, salesman must open the app on depot Wi-Fi to prime Dexie.js IndexedDB with full active product lists, prices, and merchant records.	If launched cold in a dead-zone without local cache, app warns the user to connect to cellular data to initialize local storage.
UUID ClientOrderId Idempotency	Every order created offline generates a cryptographically unique UUIDv4 clientOrderId. Salesman monitors sync status without fear of double billing.	If cellular signal drops mid-transmission, backend deduplicates by clientOrderId, guaranteeing the retailer is never double-billed.
Simulated Offline Testing Mode	The app features an interactive 'Simulated Offline' toggle in the header. Salesman can test local queue storage and test offline workflows even when cellular signal is active.	Enables hands-on field training of new salesmen without sending test orders to the warehouse picking line.
Offline Queue Inspection & Purging	Salesman opens 'Offline Queue Modal' to view pending unsynced orders, timestamps, merchant names, and amounts. Salesman can review or cancel an un-synced order before upload.	Provides complete transparency so salesmen can assure kirana owners that their order is securely recorded on the device.
Stock Depletion Conflict on Sync	If warehouse stock for an ordered item depletes while the salesman is offline in Sarafa Bazar, the backend flags the discrepancy upon synchronization.	The order is synced with a PARTIAL_STOCK_WARNING; salesman receives an alert to notify the retailer or substitute an alternative SKU.
Special Operational Notes Logging	Salesmen must log delivery constraints (e.g. *'Store in narrow alley, deliver via 2-wheeler tempo before 11 AM'* or *'Cheque ready on delivery'*).	Notes are passed verbatim into the order record, printed on picking sheets, and displayed on the Admin Order Desk.
In-Shop Khata Collection Reminders	Salesman inspects outstandingBalance. If the merchant is nearing credit exhaustion, salesman collects cash or guides the shopkeeper to scan the UPI QR code.	Prevents the salesman from booking large orders that will stall in distributor credit hold.

SECTION 5 | ROLE SPECIFICATION: DISTRIBUTOR ADMIN (ORDER DESK OPERATIONS)

Centralized Command, Risk Management, Inventory Control & Tally ERP Export

5.1 Role Overview & Operational Mandate

The **Distributor Admin / Order Desk Manager** (e.g. Ramesh Gupta, Jabalpur Central Depot) is the primary operational and financial custodian of the entire distribution network. The Admin operates the central 12-column Order Desk, monitoring an average of 600 orders daily across 1,800 retail accounts. Responsibilities span reviewing orders, mitigating credit risk, managing stock and pricing tiers, and generating compliant RFC-4180 CSV exports for Tally ERP.

5.2 Primary (Macro) Responsibilities

• **Central 12-Column Live Order Desk Operations:** Triage, inspect, and process high-volume order streams. Filter orders by status tabs: 'Pending Review', 'Credit Hold', 'Approved', and 'All Orders'. Real-time toast alerts notify admins of fresh bookings.

• **Financial Risk & Credit Hold Adjudication:** Evaluate orders flagged for exceeding merchant credit limits. Review retailer payment history and exercise authority to approve with override or reject with documented reasons.

• **FMCG Master Catalogue & Inventory Control:** Maintain master SKU definitions, packaging specs (pack sizes, units), MRP, dealer base prices, GST tax slabs (5%, 12%, 18%), and adjust physical warehouse stock counts in real time.

• **Dynamic Trade Promotion Scheme Engineering:** Configure and launch promotional schemes (Buy X Get Y Free, Percentage Bulk Discounts, Flat Discounts). Set minimum qualifying thresholds, free product mappings, and campaign validity dates.

• **Kirana Retailer CRM & Tier Assignments:** Manage retailer accounts, configure credit limits, assign price tiers (STANDARD, TIER_1, TIER_2, WHOLESALE), map shops to designated beat salesmen, and verify GSTIN tax numbers.

• **Tally ERP RFC-4180 Batch CSV Export:** Export approved order batches into standard accounting CSV templates with precise SKU codes, HSN numbers, dealer rates, tax breaks, and retailer ledger codes for zero-error Tally ERP import.

5.3 Granular Micro-Responsibilities & Subtle Operational Possibilities (Small-to-Small Details)

Operational Event / Feature	Granular Responsibility & System Behavior	Edge Case / Failure Prevention
Credit Hold Adjudication & Override	When an order exceeds credit ceiling, system flags it in amber. Admin reviews merchant khata ledger and decides whether to approve based on verbal payment promises.	If approved, an audit flag records the admin user ID and timestamp; if rejected, a mandatory rejection reason is logged and pushed to the retailer.
Order Rejection Reason Logging	Admin cannot reject an order with a blank reason. Admin must select or enter structured rejection categories: *'Credit Limit Exhausted'*, *'Damaged Batch / Stockout'*, or *'Address Outside Beat Territory'*.	The rejection reason immediately updates the order status in the retailer's app and salesman's beat list, preventing repeat ordering of bad lines.
Dispatch Pipeline Progression	Admin transitions orders through operational gates: APPROVED -> DISPATCHED (warehouse picking sheet printed) -> DELIVERED (driver proof of delivery).	Transitioning to DISPATCHED automatically decrements warehouse inventory in PostgreSQL and reserves stock against pending orders.
Tally CSV Formatting Compliance	Admin triggers batch export for Tally ERP. Admin must ensure all exported orders are in APPROVED status, with valid HSN codes, correct dealer rates, and escaped text fields.	Export engine enforces strict RFC-4180 CSV compliance: double-quotes commas, handles multi-line notes, and encodes in UTF-8 BOM to prevent Tally import crashes.
Inactive Retailer Retention Triage	Admin dashboard actively scans merchant ordering frequency and flags retailers with zero orders in the last 14 days. Admin reallocates beat routes or schedules salesman calls.	Prevents kirana attrition to competitor FMCG distributors by catching churn early.
Scheme Non-Stacking Guardrails	When creating trade promotions, Admin must verify that multiple schemes do not inadvertently stack on the same SKU (e.g. 5% bulk discount PLUS Buy 10 Get 1 Free) unless intentionally authorized.	Pricing engine applies schemes deterministically based on priority rules, protecting distributor gross margins.
Channel Origin Auditing	Admin inspects the createdVia badge on each order: PORTAL, AI_CAPTURE, SALESMAN_APP, or OFFLINE_SYNC.	Enables admin to evaluate AI adoption rates, field salesman efficiency, and track which retailers are successfully transitioning to digital ordering.

SECTION 6 | ROLE SPECIFICATION: SYSTEM & AUTOMATED ARCHITECTURAL ENGINES

The Non-Human Autonomous Actors Governing Logic, Data Integrity, and Interoperability

6.1 The System Actor: Autonomous Architectural Services

SupplySetu is powered by five mission-critical automated backend engines that act as non-human participants in the platform lifecycle. These engines execute deterministic calculations, probabilistic natural language parsing, and state synchronizations that eliminate manual office entry:

Automated Engine / Service	Core Execution Mandate & Logic Rules	Safety Guardrails & Fallbacks
Gemini AI Order Extraction Engine	Ingests unstructured Hinglish/Hindi text notes from WhatsApp or voice transcripts. Uses Google Gemini API (@google/genai) to extract SKU codes, quantities, and packaging units.	**Phonetic Fuzzy Fallback:** If AI API is unreachable or ambiguous, backend falls back to local Levenshtein/regex matching. Output is always presented to the user for confirmation.
Dynamic Tier Pricing Engine	Computes invoice line items: BasePrice * (1 - TierDiscount), then adds applicable GST (5%, 12%, 18%). Evaluates dealer margins vs consumer MRP.	**Server-Side Determinism:** Client-submitted prices are discarded; backend recalculates all line items against database rate cards to eliminate client tampering.
Automated Trade Scheme Engine	Evaluates order basket against active schemes. Calculates BUY_X_GET_Y_FREE bonus units using integer floor division; calculates percentage discounts on qualifying volumes.	Free products are injected as Rs 0.00 line items with tax-exempt treatment according to Indian GST trade promotional rules.
Dexie IndexedDB Sync Worker	Monitors navigator online status. Automatically drains offline order queue upon reconnection, executing atomic POST requests with UUID idempotency keys.	**Exponential Backoff & Dedup:** Prevents duplicate submissions on unstable connections and stores failed payloads for manual retry.
Live Notification & SSE Hub	Dispatches Server-Sent Events (SSE) and WebSocket alerts to connected Admin Order Desks when new orders arrive, accompanied by audio toasts and metric refreshes.	Guarantees that warehouse pickers and office clerks receive instant visibility without manual browser page reloading.

6.2 Technical Failure Recovery & Exception Handling Protocols

Failure Condition / Edge Case	Autonomous System Response	Human Operator Action
Gemini AI API Network Timeout	System engages local fuzzy regex parser; maps obvious keywords (e.g. 'parle g', 'surf excel') and flags unparsed lines.	Retailer manually verifies unmapped items on interactive review screen.
Simultaneous Out-of-Stock Race	Prisma transaction fails on order approval if stock decrements below zero; order is automatically placed on stock hold.	Admin contacts retailer to propose alternate pack size or partial delivery.
Network Flicker During Offline Sync	Sync worker traps HTTP failure, retains order in Dexie IndexedDB queue, and retries in 15 seconds using exponential backoff.	Salesman sees amber 'Syncing' badge; no manual data re-entry required.

SECTION 7 | CROSS-ROLE WORKFLOWS & END-TO-END OPERATIONAL SCENARIOS

Step-by-Step Multi-Role Interaction Lifecycles Across Real-World Distribution Scenarios

The power of SupplySetu lies in how these roles interact seamlessly across complete operational cycles. The following four real-world scenarios trace multi-role handoffs from initial demand capture through warehouse dispatch and accounting export:

SCENARIO A: The Messy WhatsApp Natural Language AI Order Flow

- **Step 1 (Ingestion):** Kirana retailer receives a messy list on WhatsApp from shop assistant: *'bhaiya 10 peti parle g, 4 peti surf excel 1kg, 2 peti red label'*. Retailer copies text into SupplySetu AI Order Pad.
- **Step 2 (AI Parsing):** Gemini AI parses text, resolving 'parle g' to PARLE-G-80G (10 boxes), 'surf excel 1kg' to SURF-EXCEL-1KG (4 boxes), and 'red label' to RED-LABEL-TEA-250G (2 boxes).
- **Step 3 (Human Review):** The review modal opens. Retailer sees Red Label Tea is out of stock in depot; replaces with Tata Tea. Verifies Parle-G triggered 'Buy 10 Get 1 Free' scheme (+1 free box). Confirms cart.
- **Step 4 (Admin Triage):** Order lands on Admin Order Desk tagged as createdVia: AI_CAPTURE. Admin reviews line items, sees retailer has adequate credit, and clicks 'Approve Order'.
- **Step 5 (ERP Export):** Approved order is included in the evening Tally ERP batch export, ready for accounting reconciliation.

SCENARIO B: The Remote Beat Zero-Network Offline Punch & Sync

- **Step 1 (Offline Arrival):** Beat salesman arrives at Sarafa Bazar shop where mobile connectivity drops completely. App header switches to offline badge.
- **Step 2 (Local Booking):** Salesman opens shop profile, selects 15 cases of detergent, and punches order. Order is written to local Dexie.js IndexedDB with UUID clientOrderId: c7b3.... Salesman receives local confirmation.
- **Step 3 (Queue Storage):** Salesman checks 'Offline Queue'; sees the pending order timestamped and securely cached.
- **Step 4 (Sync Trigger):** Salesman drives out of bazaar into 4G zone. The Sync Engine detects connection, initiates background upload, and clears the Dexie queue upon HTTP 201 receipt.
- **Step 5 (Depot Alert):** Live sound alert sounds at distributor depot; order appears on live desk tagged createdVia: OFFLINE_SYNC.

SCENARIO C: Credit Ceiling Exceedance & Instant QR Resolution

- **Step 1 (Limit Exceedance):** Retailer with Rs 75,000 credit ceiling and Rs 62,000 existing debt attempts to book a Rs 20,000 stock order. Total liability = Rs 82,000 (exceeds limit by Rs 7,000).
- **Step 2 (Credit Hold):** The system automatically places the order in CREDIT_HOLD status instead of clean approval.
- **Step 3 (Admin Review):** Admin Order Desk receives the order in the 'Credit Hold' tab with amber highlighting. Admin inspects merchant khata history and contacts shopkeeper.
- **Step 4 (Instant Payment):** Retailer opens Khata tab, clicks 'Pay via UPI', inputs Rs 10,000 payment, scans QR, and submits. Balance drops to Rs 52,000.
- **Step 5 (Order Release):** Admin refreshes desk, verifies credit headroom is now Rs 23,000, and immediately approves the order for warehouse dispatch.

SCENARIO D: Dormant Kirana Churn Detection & Field Recovery

- **Step 1 (Churn Scan):** Analytics Engine detects Shri Balaji Store has not placed an order in 16 days (exceeds 14-day threshold).
- **Step 2 (Admin Alert):** Admin overview dashboard flags Balaji Store under 'Inactive Retailers' with priority red badge.
- **Step 3 (Salesman Dispatch):** Admin re-routes Beat #4 salesman Vikram Sharma to prioritize an in-person retention visit.
- **Step 4 (Retention Order):** Salesman visits store, offers 3% volume promotional scheme on tea leaves, and punches an on-behalf order.

SECTION 8 | RACI GOVERNANCE MATRIX & ENTERPRISE SIGN-OFF

Responsibility Assignment, Security Protocols, and Production Verification Standard

8.1 Complete RACI Responsibility Assignment Matrix

The RACI matrix below governs all core operational actions (**R** = Responsible, **A** = Accountable, **C** = Consulted, **I** = Informed):

Platform Capability / Operational Task	Retailer	Salesman	Admin	System / AI
Self-Service Digital Order Creation	**R / A**	I	I	C (Validates)
On-Behalf Beat Order Booking	C (Approves)	**R / A**	I	C (Calculates)
Unstructured Text / Hinglish AI Parsing	C (Reviews)	C	I	**R / A (Parses)**
Offline Order Caching & Sync Execution	I	**R**	I	**A (Atomic Sync)**
Credit Limit Overrun Adjudication	I	C	**R / A (Decides)**	R (Flags Hold)
Master SKU Base Price & MRP Setup	I	I	**R / A**	C
Trade Scheme Formulation & Activation	I	I	**R / A**	R (Applies)
Tally ERP RFC-4180 Batch CSV Generation	N/A	N/A	**R / A**	R (Formats)
Khata Debt Settlement & UPI Payment	**R / A**	C (Follows Up)	I (Audits)	R (Credits)
Inactive Kirana Churn Detection & Re-Route	I	R (Re-Visits)	**A (Re-Routes)**	R (Scans & Flags)

8.2 Security Boundaries, Access Control & Data Protection Mandates

• **Database Multi-Tenant Isolation:** All retailer queries in Prisma ORM strictly scope by where: { retailerId: user.retailer.id }. Retailers cannot query or mutate records outside their shop foreign keys. Salesmen can only access merchants mapped in their assigned beat table.

• **Strict Server-Side Price Authority:** The frontend never dictates pricing or taxes. Any prices sent from the client are disregarded; the server reconstructs cart pricing using the retailer's assigned PriceTier and the current Product master table.

• **Idempotent Offline Guarantees:** Offline orders use client-generated UUID keys (clientOrderId). In case of duplicate HTTP POST attempts due to cellular retry storms, the backend enforces uniqueness constraints, returning the existing order record without double-debiting inventory.

• **Tally RFC-4180 CSV Hardening:** All exported fields are sanitized against formula injection (stripping leading '=', '+', '-'), enclosed in RFC-4180 quotes, and encoded in UTF-8 BOM to prevent corrupt imports into legacy Tally ERP versions.

OPERATIONAL CONCLUSION & SYSTEM SIGN-OFF: By enforcing deterministic roles, unambiguous boundaries, and automated safety fallbacks, SupplySetu eliminates over 95% of manual ordering errors, reduces order processing latency from 18 hours to under 30 minutes, and secures distributor cash flow across extensive FMCG distribution networks.